Should a vegan eat higher-welfare meat? A quick model

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1 Preface

In this, I introduce an elementary partial-equilibrium framework for whether a vegan should eat higher-welfare meat (or eggs) that is still net negative for animal welfare, assuming linear supply and demand curves.

2 Set-up

Suppose there are two retail markets, one for higher-welfare meat (indexed by H) and one for lower-welfare meat (indexed by L). Let P_H and P_L denote prices and Q_H and Q_L denote quantities. For simplicity, I assume that demand and supply are linear in prices, with common own-price slopes $\beta > 0$ in demand and $\epsilon > 0$ in supply, and with common cross-price slopes $\gamma \geq 0$ in demand and $\phi \geq 0$ in supply. Higher-welfare meat is costlier to produce, so we have a per-unit "wedge" $c \geq 0$; producers therefore respond to the net margin $P_H - c$ when making this allocation decision. The constants α_H, α_L, and δ are intercepts. Hence, the system of equations is

$$Q_H^d = \alpha_H - \beta P_H + \gamma P_L, \tag{1}$$
$$Q_L^d = \alpha_L - \beta P_L + \gamma P_H, \tag{2}$$
$$Q_H^s = -\delta + \epsilon(P_H - c) - \phi P_L, \tag{3}$$
$$Q_L^s = -\delta + \epsilon P_L - \phi(P_H - c). \tag{4}$$

I'll assume $\beta + \epsilon > \gamma + \phi$, which ensures that own-market price responses dominate cross-market substitution (I'll call this "own-market dominance"); this ensures that the system of equations has a positive determinant.

Let w_H and w_L denote per-animal welfare in higher-welfare and lower-welfare production respectively. I assume that both are negative, but that lower-welfare conditions are strictly worse, so $w_L < w_H < 0$. Total animal welfare is given by $W = w_H Q_H + w_L Q_L$. Define the welfare ratio $r \equiv |w_L|/|w_H| > 1$, which measures how much worse lower-welfare conditions are per animal than higher-welfare conditions. Without loss of generality, we can set $w_H = -1$ and $w_L = -r$.

3 Comparative statics

The market-clearing conditions $Q_H^d = Q_H^s$ and $Q_L^d = Q_L^s$ can be rearranged to

$$(\beta + \epsilon)P_H - (\gamma + \phi)P_L = \alpha_H + \delta + \epsilon c, \tag{5}$$
$$(\beta + \epsilon)P_L - (\gamma + \phi)P_H = \alpha_L + \delta - \phi c. \tag{6}$$

Consider a vegan deciding to eat a unit of higher-welfare meat, i.e., a unit upward shift in demand for higher-welfare meat, modeled as $\Delta\alpha_H = 1$ with α_L and c held fixed. Differentiating the system of equations gives

$$1 = (\beta + \epsilon)\,\Delta P_H - (\gamma + \phi)\,\Delta P_L, \tag{7}$$

$$0 = (\beta + \epsilon)\,\Delta P_L - (\gamma + \phi)\,\Delta P_H. \tag{8}$$

Defining $D \equiv (\beta + \epsilon)^2 - (\gamma + \phi)^2$, which is strictly positive thanks to the assumption that $\beta + \epsilon > \gamma + \phi$, the price changes are

$$\Delta P_H = \frac{\beta + \epsilon}{D}, \qquad \Delta P_L = \frac{\gamma + \phi}{D}.$$

Substituting these into the supply equations, we have

$$\Delta Q_H = \epsilon\,\Delta P_H - \phi\,\Delta P_L = \frac{\epsilon\beta + \epsilon^2 - \phi\gamma - \phi^2}{D}, \tag{9}$$

$$\Delta Q_L = \epsilon\,\Delta P_L - \phi\,\Delta P_H = \frac{\epsilon\gamma - \phi\beta}{D}. \tag{10}$$

4 The (animal) welfare condition

The change in total animal welfare induced by the marginal demand shift is

$$\Delta W = w_H\,\Delta Q_H + w_L\,\Delta Q_L = -\Delta Q_H - r\,\Delta Q_L.$$

Substituting the above expressions, we get

$$\Delta W = -\,\frac{(\epsilon\beta + \epsilon^2 - \phi\gamma - \phi^2) - r\,(\phi\beta - \epsilon\gamma)}{D}.$$

Since $D > 0$, the sign of ΔW is determined by the sign of the numerator. Hence, the marginal demand shift (by the once-vegan) toward higher-welfare meat strictly increases total animal welfare if and only if

$$(\epsilon\beta + \epsilon^2 - \phi\gamma - \phi^2) - r(\phi\beta - \epsilon\gamma) < 0.$$

Thus, when $\phi\beta > \epsilon\gamma$, the marginal higher-welfare demand increase increases total animal welfare if and only if

$$r > \frac{\epsilon\beta + \epsilon^2 - \phi\gamma - \phi^2}{\phi\beta - \epsilon\gamma}.$$

Meanwhile, when $\phi\beta < \epsilon\gamma$, dividing by $\phi\beta - \epsilon\gamma < 0$ reverses the inequality, so the welfare condition becomes

$$r\frac{\epsilon\beta + \epsilon^2 - \phi\gamma - \phi^2}{\phi\beta - \epsilon\gamma}.$$

But this cannot hold for any $r > 1$: combining $\phi\beta < \epsilon\gamma$ with own-market dominance forces $\epsilon > \phi$ (otherwise $\phi\beta < \epsilon\gamma \leq \phi\gamma$ would give $\beta < \gamma$, contradicting $\beta + \epsilon > \gamma + \phi$), hence $\epsilon(\beta + \epsilon) > \phi(\beta + \epsilon) > \phi(\gamma + \phi)$; the numerator is therefore positive while the denominator is negative, so the right-hand side is negative.

Hence, the two conditions that must *both* be met for this marginal demand increase to be welfare positive are

$$\boxed{\begin{gathered} \phi\beta > \epsilon\gamma, \\ r > \frac{\epsilon\beta + \epsilon^2 - \phi\gamma - \phi^2}{\phi\beta - \epsilon\gamma}. \end{gathered}}$$

Since $\Delta Q_L = (\epsilon\gamma - \phi\beta)/D$, the condition $\phi\beta > \epsilon\gamma$ is exactly the requirement that the demand shift causes lower-welfare output to fall ($\Delta Q_L < 0$). Without it, the marginal vegan adds animals to the system on net. With it, supply-side substitution actually pulls some animals out of the worse system and into the better one. The second condition then asks that the welfare ratio r be large enough that the welfare gained from those displaced lower-welfare animals outweighs the welfare cost of the additional higher-welfare animals brought into existence ($\Delta Q_H > 0$, the normal own-market response).

In practice, I'd guess these conditions are hard to meet, but I'm not certain.